

People's Democratic Republic of Algeria

Kasdi University Merbah Ouargla
Computer Architecture
TOUMI Chahrazad

TD n° 4

Exercise 1

Complete the following table (to four decimal places).

Decimal		11,625				12.900	
Binaire	11,011						11101,11
Octal			17,13		12.700		
Hexadecimal				A0,F			

Exercise 2

Find the results of the following arithmetic operations:

- 1) $11,01_2 + 101,1_2$
- 2) $110100,1_2 + 101,01_2$
- 3) $1001,0101_2 - 0100,1010_2$
- 4) $11001_2 - 0100,1010_2$
- 5) $10011,11_2 \times 110100_2$
- 6) $10000,111_2 \times 1011,10_2$
- 7) $1011,01_2 / 100_2$
- 8) $10100001_2 / 111_2$
- 9) $1111010_2 / 1010_2$
- 10) $111110100100_2 / 10101011_2$

Exercise 3

Provide the representation of the following numbers using the IEEE 754 standard in both single and double precision: $10011,11_2$, $35,5_{10}$, $-74,0_8$, $-0,007_8$, $+0$, -0 , 2 , $+\infty$, 255_{10} , 128_{10}

Now, provide the decimal value of the following numbers represented using the IEEE 754 standard single precision: FF800000, 7FC00000, BE400000, 3F800000, 80000000, 7FC00001, 3FC00000